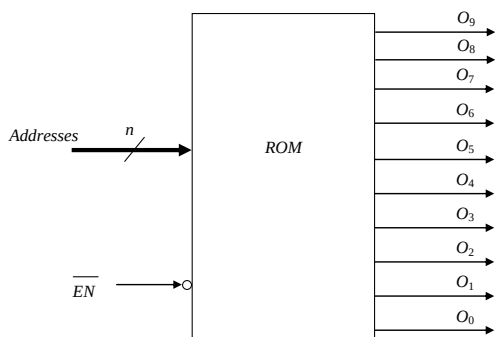


Subject LD3, January 2021

1. (20 minutes) a) How can the Quine-McCluskey algorithm be applied for minimizing Boolean functions in the conjunctive canonical form?
- b) Apply this algorithm on the following Boolean function: $f = \prod(0,1,5,12,14) + \prod_{\phi}(3,4,7,8,10,13)$
- c) Express f in the conjunctive canonical form and in the disjunctive canonical form.

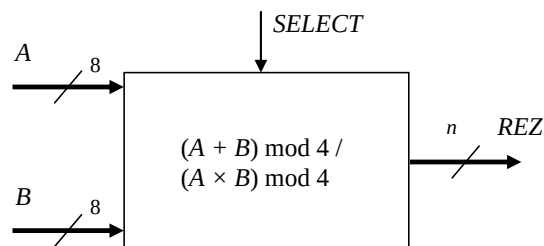
2. (20 minutes) The schematic below presents a ROM chip whose capacity is of 10,240 bits. When the EN input is active, the chip is selected and works as a regular ROM module. When EN input is not active, the chip is not selected and the outputs are in high impedance state.

- a) Determine the value of n (the width of the Address bus). Justify your answer.
- b) Show how to combine two of these chips to produce a $2\text{kb} \times 8$ memory module. The circuit should use no other logic devices.



3. (20 minutes) Design with MSI and/or SSI components a universal 4-bit device with the following functioning regimes: a) synchronous parallel load; b) synchronous shift right by one bit; c) synchronous count-up; d) synchronous shift right by two bits; e) asynchronous reset.

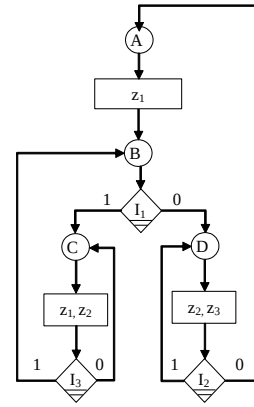
4. (40 minutes) Design using SSI components a digital system whose inputs are two 8-bit numbers, A and B , expressed in the *Unsigned Integers* numbering system, and whose output represents either their modulo 4 sum, or their modulo 4 product, according to a *SELECT* signal. The output is called REZ; this is an n -bit number – specify the value of n .



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5. (20 minutes) Design the sequential logic system described by the nearby state diagram, using any method you prefer. Explain why you prefer the chose method.

- Is it a Moore machine or a Mealy machine? Justify your answer.
- Draw the black box of this system.
- Apply the synthesis method and outline all the necessary design steps.



6. (30 minutes) Design with SSI components a Full Adder that is able to add an 8-bit number A to a 4-bit number B with CarryIn, CarryOut and Overflow Detection. Explain your answer (i.e. the design decisions).

7. (30 minutes) Design with D Flip-Flops a reversible counter whose counting loop contains the following states: 0, 0, 4, 5, 11, 12, 11, 14, 15, 3, and then back to the beginning. Show all the steps of the synthesis method.